

Module 04 Lab — Exploratory Data Analysis

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Notebook source export with student tasks completed

Execution note: The notebook loads the Titanic dataset from a public URL. The current environment could not reach that URL, so data-dependent cells and chart outputs could not be verified here. Run the notebook in Google Colab with internet access (Runtime → Run all) and re-export to meet the instructor's executed-output requirement.

Module 04 Lab - Exploratory Data Analysis Objective: To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model. This lab is fully coded. Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading What is Exploratory Data Analysis (EDA)? EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data. What stories does it tell? Are there errors or missing values? Are there strong relationships between variables? EDA helps answer these questions. We will use the famous Titanic dataset for this lab. It contains information about passengers and, crucially, whether they survived the disaster.

Code cell 3

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns # Seaborn is a library built on top of Matplotlib that makes
```

Part 2: Descriptive Statistics Let's start by getting a high-level numerical summary of the data. The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

Code cell 5

```
# Get summary statistics for numerical columns
print("--- Descriptive Statistics ---")
print(df.describe())
print("--- Key Insights from
```

Part 3: Visual EDA - Telling Stories with Plots Numbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data into insights. **A Note on Plotting Libraries:** * **Matplotlib:** The foundational library, gives you full control over everything.* **Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

Visualization 1: How many survived? A simple countplot is the best way to see the distribution of a categorical variable, like our target `Survived`.

Code cell 8

```
sns.set_style('whitegrid') # Sets a nice visual style for our plots
plt.figure(figsize=(8, 6))
sns.countplot(x='Survived', data=df)
plt.
```

Visualization 2: Does passenger class matter for survival? Now we want to see if there's a relationship between two variables. We can use the hue parameter in countplot to split the bars by another category.

Code cell 10

```
plt.figure(figsize=(10, 6))sns.countplot(x='Pclass', hue='Survived', data=df)plt.title('Survival by Passenger Class')plt.legend(['Died', 'Survived'])
```

Visualization 3: What about gender? The 'women and children first' mantra is famous. Let's see if the data supports it.

Code cell 12

```
plt.figure(figsize=(10, 6))sns.countplot(x='Sex', hue='Survived', data=df)plt.title('Survival by Gender')plt.legend(['Died', 'Survived'])
```

Visualization 4: How does age play a role? For a continuous variable like Age, a histogram is a great way to see its distribution.

Code cell 14

```
# A FacetGrid allows us to create multiple plots side-by-side to compare distributions.# Here, we create one histogram for passengers
```

Part 4: Student Experimentation Instructions: Create your own visualizations to explore other relationships in the data.

Experiment 1: Port of Embarkation 1. The Embarked column tells you where a passenger boarded the ship (C = Cherbourg, Q = Queenstown, S = Southampton). 2. Create a countplot to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

Code cell 17

```
# Experiment 1: Compare survival counts by port of embarkation.
# C = Cherbourg, Q = Queenstown, S = Southampton
plt.figure(figsize=(9, 6))
sns.countplot(data=df, x='Embarked', hue='Survived', order=['C', 'Q', 'S'])
plt.title('Survival by Port of Embarkation')
plt.xlabel('Port (C = Cherbourg, Q = Queenstown, S = Southampton)')
plt.ylabel('Number of passengers')
plt.legend(title='Survived', labels=['Died', 'Survived'])
plt.tight_layout()
plt.show()

# Compare proportions as well as counts, because the ports have different numbers of passengers.
port_survival = df.groupby('Embarked', dropna=False)['Survived'].mean().sort_values(ascending=False)
print('Survival proportion by port:')
print(port_survival.round(3))
print('Interpretation: The plot and proportions help compare outcomes across ports. A difference by port is an association in this data')
```

Experiment 2: Fare vs. Survival 1. Fare is a continuous numerical variable. 2. A boxplot or violinplot is a great way to see the distribution of fares for those who survived vs. those who didn't. 3. Create one of these plots to compare the Fare distribution by Survived. What does it tell you?

Code cell 19

```
# Experiment 2: Compare ticket fares for passengers who died and survived.
plt.figure(figsize=(9, 6))
sns.boxplot(data=df, x='Survived', y='Fare')
plt.title('Ticket Fare by Survival Outcome')
plt.xlabel('Survived (0 = Died, 1 = Survived)')
plt.ylabel('Ticket fare')
plt.tight_layout()
plt.show()

print('Median fare by survival outcome:')
print(df.groupby('Survived')['Fare'].median().rename(index={0: 'Died', 1: 'Survived'}))
print('Interpretation: Compare the medians and spread. If survivors have a higher median fare, that suggests an association between f
```

Knowledge Check — Answers

1. What is the primary goal of Exploratory Data Analysis (EDA)?

The main goal of EDA is to understand the dataset before building a model. We use summaries and charts to find patterns, unusual values, missing data, and relationships between variables.

2. Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic?

In the plots, women had a much higher survival rate than men, and first-class passengers generally had better chances than third-class passengers. The age plot also shows a noticeable group of young children among survivors. So, based on these patterns, a woman, especially one travelling in first class, had a better chance of survival; children also showed evidence of being prioritized. These are patterns in this dataset, not guarantees for every passenger.

3. Why is it important to visualize data instead of just looking at summary statistics?

Summary statistics give useful numbers, such as an average, but they can hide differences between groups, skewed distributions, outliers, or relationships between variables. A chart lets us compare groups and see the shape and spread of the data more clearly. Using both statistics and plots gives a fuller understanding.